DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PUNJABI UNIVERSITY, PATIALA

COMPUTER SYSTEM ARCHITECTURE

2nd Year

MST-2

Section-A (Do All)

MM 15

Q1)	What is the difference between SRAM and DRAM?	(2)
Q2)	What is pipelining?	(1)
Q3)	What is locality	(2)

Section-B (Any two)

Q5)	What is locality of reference? Explain its types? How this principle is used in cache memory?	(5)
Q6)	Explain various hazards in instruction pipeline?	(5)
Q7)	Discuss DMA with its block diagram?	(5)

Submit your MST at:

jagroop_80@csepup@ac.in for CE12 and CE34

Kahlon.navroz3@gmail.com for CE56 and CE78

B Tech 2nd year

MM:15 **Computer Architecture** TT:1 hour

Section A is compulsory. Attempt any two from section B.

Section A

Q1

a) What is memory mapped I/O? (1)

b) Write about bootstrap loader. (1)

c) Explain hit ratio of memory. (1)

d) Briefly explain Input Output processor (2)

Section B

Q2 Explain the DMA mode of transfer with detailed diagrams. (5)

Q3 Write about following: (5)

i) Fault page condition.

ii) Locality of Reference

iii) Mapping process

Q4 Explain interrupt cycle of basic computer system. (5)

Subject – Computer System Architecture

MST – 2

Time: 1 Hr. M. Marks: 15

Note: Section – A is compulsory. Do any two questions from Section – B.

Section – A

Q1. What are the representations of integer number (-19) in an 8-bit format using:

a) Signed Magnitude method

b) Signed 2's Complement method (2)

Q2. Compute 0.625 + (- 0.75) using fixed point numbers. (2)

Q3. Differentiate between RAM and ROM. (1)

Section – B

Q1. What is cache memory? How does the performance of cache memory be measured? (2+3)

Q2. Multiply (33) and (-4) using Booth's Algorithm and show the contents of registers AC, Q for the process of multiplication. (5)

Q3. Discuss the methods used for expansion of memory. Draw the diagram to obtain a 1024 x 8 memory using 128 x 8 memory chips. (2+3)

Department of Computer Science and Engg

Computer System Architecture

TT 1 Hr MM 15

Section A

Q:1 a. What is cache coherence?

b. What is Control memory?

c. What are pipeline hazards?

d. What is cycle stealing in DMA.

e. Explain micro instruction format. 5

Section B(Do any two)

Q:2 Explain the process of pipeline in detail. 5

Q:3 Design micro program sequencer and discuss in detail. 5

Q:4 Explain the working of DMA. 5